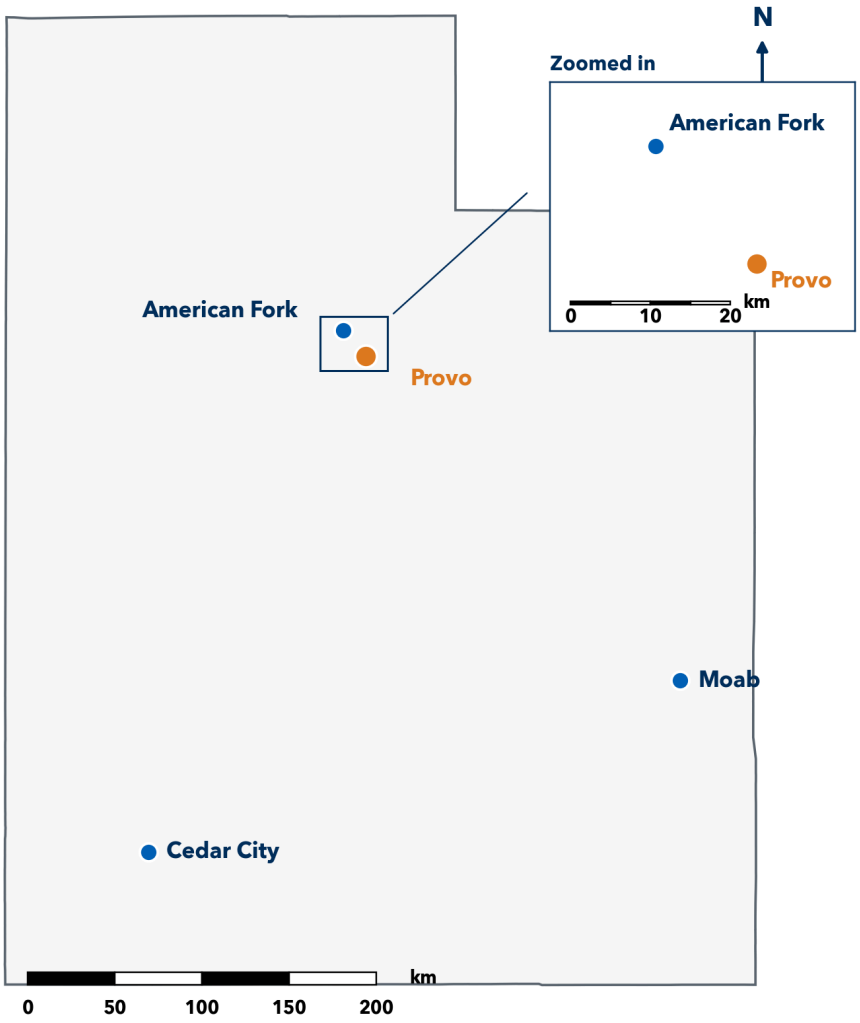
What Is the Temperature in Provo?

CE 414 · Week 9 · Sampling and Interpolation · in-class worksheet, done on your own

Name: _____ Date and time: _____



1. Gather the samples

Look up the **current air temperature** in each city (weather.com or similar). Then find each city's **straight-line distance to Provo**: measure it on the map with the scale bar (use the zoomed-in box for American Fork), or use the measure tool in an online map.

City	Temperature (°F)	Straight-line distance to Provo (km)
American Fork		
Moab		
Cedar City		

Source of the temperatures: _____

2. Your own method

Using only the three temperatures above, estimate Provo's temperature any way you like.

Provo estimate: _____ °F

How did you get it?

3. Thiessen polygons (nearest neighbor)

On the map, draw a line joining each pair of known cities, then the **perpendicular bisector** of each line. Join the bisectors into three polygons. Provo takes the value of the polygon it falls in.

Provo falls in the polygon of: _____ **Provo estimate:** _____ °F

4. Nearest neighbors, n = 2

Average the temperatures of the **two cities closest** to Provo.

Two closest: _____ (_____ + _____) ÷ 2 = **Provo:** _____ °F

5. Inverse distance weighted (IDW), power 2

Weight each city by **1 / d²**, so closer cities count more, and take the weighted average of all three.

$$\text{Provo} = \Sigma\,(w \cdot T) / \Sigma\, w, \quad w = 1/d^2$$

City	T (°F)	d (km)	w = 1 / d²	w · T
American Fork				
Moab				
Cedar City				
Sum				

Provo estimate = Σ(w · T) ÷ Σw = _____ °F

Tip: the weights are tiny numbers. Keep at least three significant figures, or work in 1 / (d ÷ 10)² instead: scaling every weight by the same amount does not change the answer.

6. Check

Provo's actual temperature right now: _____ °F

Which method came closest, and why do you think it did? Which would you trust if Provo had no thermometer?